clude bank frauds, structural defects, medical problems or errors in text. Image analysis by Uncanny Vision is an example. The system analyses the images through a CCTV and points out anomalies, with basics going down to a person falling.

Proper hardware support is also necessary

Neural networks have existed since the early 1990s, but initially success was very limited due to the restricted availability of hardware and inadequate learning methods. “Last decade has been a welcome change in this regard,” says Dr Reger. He adds, “Advanced learning algorithms and massive 16-bit FP performance in modern hardware have turned neural networks into an effective technology for image analysis.”

Movidius Myriad 2, Eyeriss and Microsoft HoloLens are some vision microprocessors being used in today’s vision-processing systems. These may include direct interfaces to take data from cameras with a greater emphasis on on-chip dataflow. These are different than regular video-processing units as these are suited more to running machine vision algorithms such as CNNs and scale-invariant feature transform (SIFT). Some of the application areas for vision processing include robotics, the IoT, digital cameras for virtual reality and augmented reality, and smartcameras.

Fooling an AI

Reports released last year suggested that changing some pixels in a photo of an elephant could fool a neural net into thinking it is a car. AI systems have committed some disturbing tasks as well. April Taylor from iTech Post has explained in an article how her dogs were categorised as horses.

Twitter user, jackyalcine, has also reported some funny business with Google AI. This has lead to Google’s chief social architect, Yonatan Zunger, releasing an apology and attempts to rectify the error, resulting in the removal of the tag altogether. The explanation behind the error came to light once it was released for public.

The problem with Google AI was training using animal images. Training an AI is a process to determine a large set of parameters that calibrates a function that maps image data to content. Results may vary based on the images used in training. Some sample images tested by enthusiasts gave some very creepy results. Even a relatively simple neural network over-interpreted an image, resulting in trippy images.

Engineers at Google’s research lab decided to check on patterns the system recognised and set up a system that tweaks the patterns to exaggerate the result. An image passed through enough times resulted in the image changing radically, and basically tripping some classy paintings.

How it looks

“After logic programming with languages like Prolog and expert systems, the present phase of advanced neural networks constitutes the third wave of AI,” says Dr Reger. But we are still decades away from a system that can get anywhere close to a realistic one. This brilliance of our human body is the reason why researchers have been trying to break the enigma of computer vision by analysing the visual mechanics of human beings or other animals.

Background clutter, hidden parts of images, difference in illumination and variations in viewpoint are some areas that require focus in development before we can move towards developing Jarvis. According to Dr Reger, “Neural networks in general and image analysis in particular will remain some of the most interesting and important topics in cutting-edge information technology in years to come.” EFY